Himalayan Mountaineering Expeditions

EXECUTIVE SUMMARY

Using “The Himalayan Database”(TidyTuesday [1]) , containing hundreds of entries on mountain ranges and expeditions, this project would explore difficulties encountered during Himalayan mountaineering expeditions.

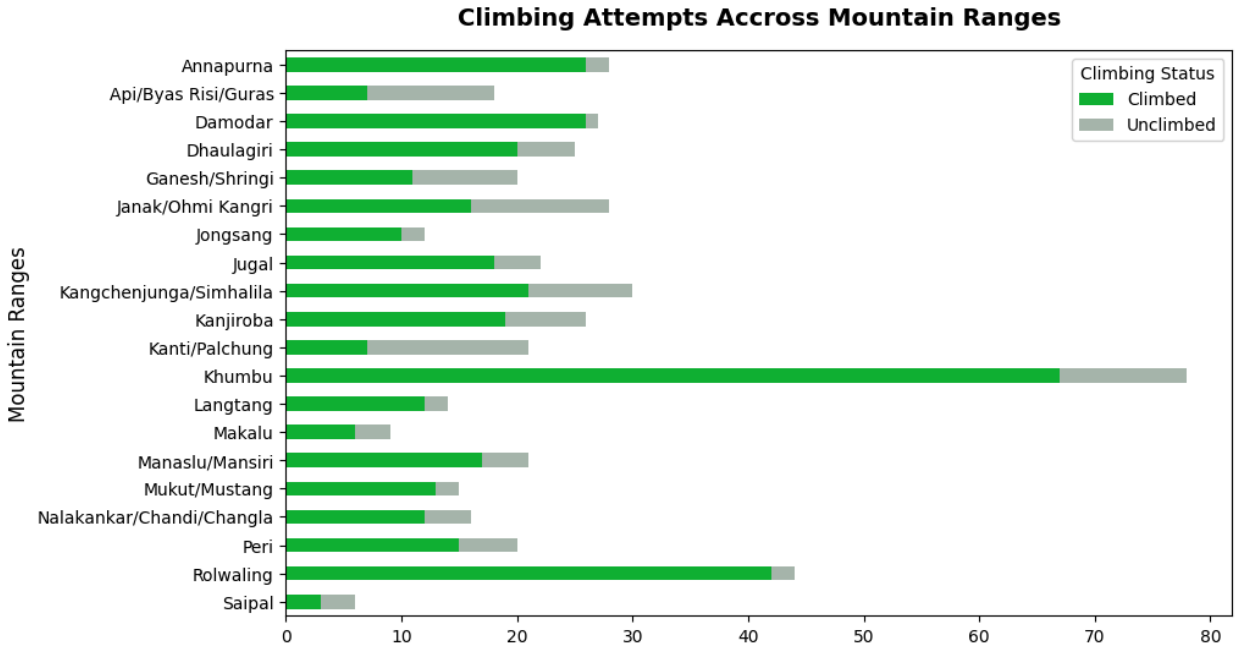
OVERVIEW

Climbing mountains is a way humans challenge their biological limits, and the Himalayas contain some of the highest mountains on Earth, including mount Everest. Climbing teams face extreme weather conditions, temperatures, and lower oxygen levels. This project uses “The Himalayan Database”(TidyTuesday [1]) to explore some of the peaks and factors affecting climbing. This database has expedition information ranging from 2020 to 2024, and has 400+ entries on peak information and 800+ on expeditions.

ANALYSIS

What is the distribution of climbing status (PSTATUS) across different mountain ranges(HIMAL_FACTOR)?

For gaining basic understanding, a good place to start is from climbing status. From viewing the distribution of climbing status, an overall understanding of how popular and difficult a mountain is for climbing can be gained. Mountain ranges such as Khumbu are climbed much more than others,suggesting popularity amongst the mountain climbing community. Api/Byas Risi/Guras have the highest amount of unclimbed over total number of attempts, suggesting a high difficulty in summiting.



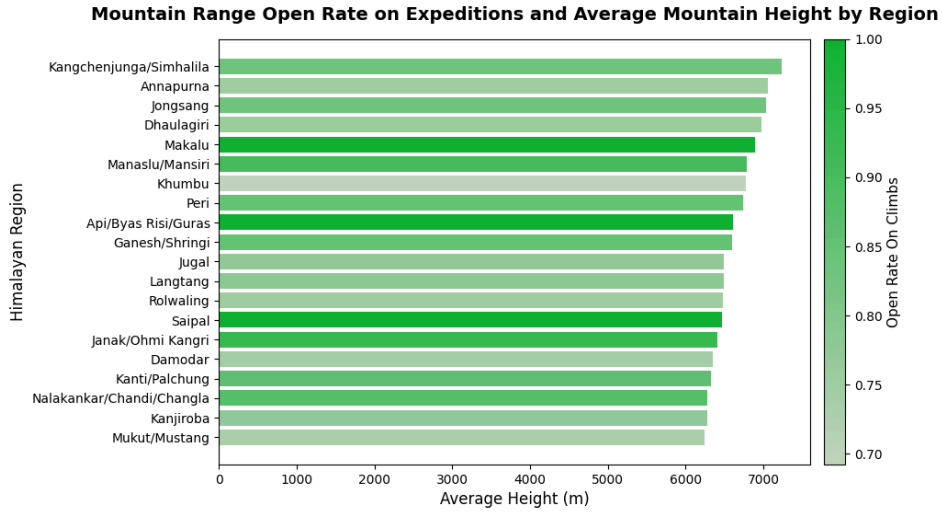
Which mountain range (HIMAL FACTOR) has the highest average peak height (HEIGHTM)?

One might think the height of a mountain would play a large factor in the popularity or unclimbed rate of a mountain range. However a quick calculation of top five mountains with highest average peak heights would show there is no obvious correlation. This suggests that weather and terrain would play a more major factor in the success rate of reaching the summit.

Mountain Peak	Height (Meters)
Kangchenjunga/Simhalila	7238
Annapurna	7058
Jongsang	7032
Dhaulagiri	6981
Makalu	6901

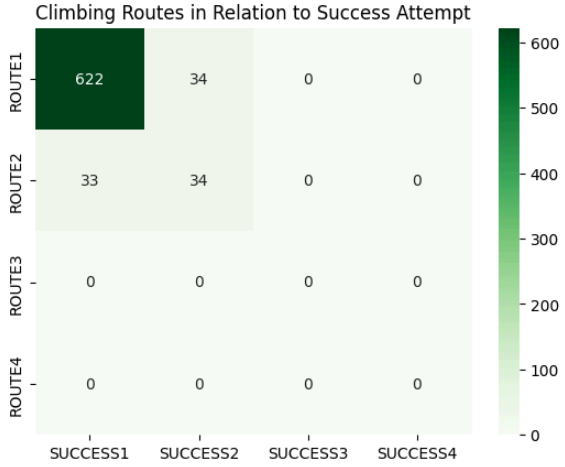
What is the distribution of peak heights (HEIGHTM) for peaks that are open (OPEN) versus those that are not?

Mountain ranges are usually closed during seasons with dangerous weather, which is a factor in difficulty in climbing a mountain. It would be natural to think the height of the mountain would likely mean less times opened. However there is no obvious pattern in whether the mountain is open during an expedition and its height. Although it is interesting to note almost all mountains have an open rate higher than 0.7. This might also be affected by expedition teams that are inexperienced and choose to ignore mountain open times.



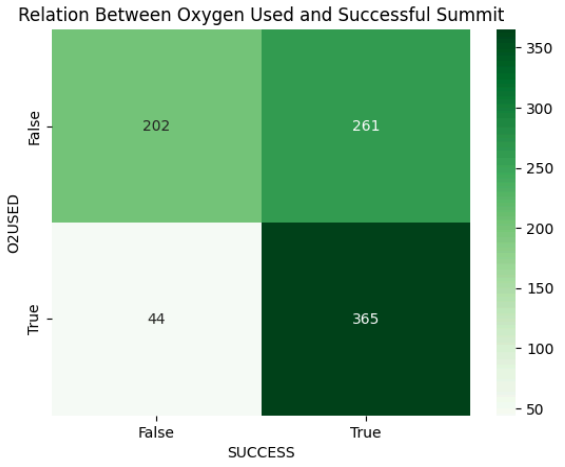
Which climbing routes (ROUTE1, ROUTE2, ROUTE3, ROUTE4) have the highest success rates (SUCCESS1, SUCCESS2, SUCCESS3, SUCCESS4) across all expeditions?

Sometimes it might take multiple summit attempts due to changing weather conditions and other possible factors. Route one has a success rate of 0.28 compared to o.21 for route 2. There is a significant difference in entry numbers for these two routes, and no data for route three and four. Most expeditions succeed on their first attempt, and there are no expeditions during this time period that succeeded on their third or fourth attempt.



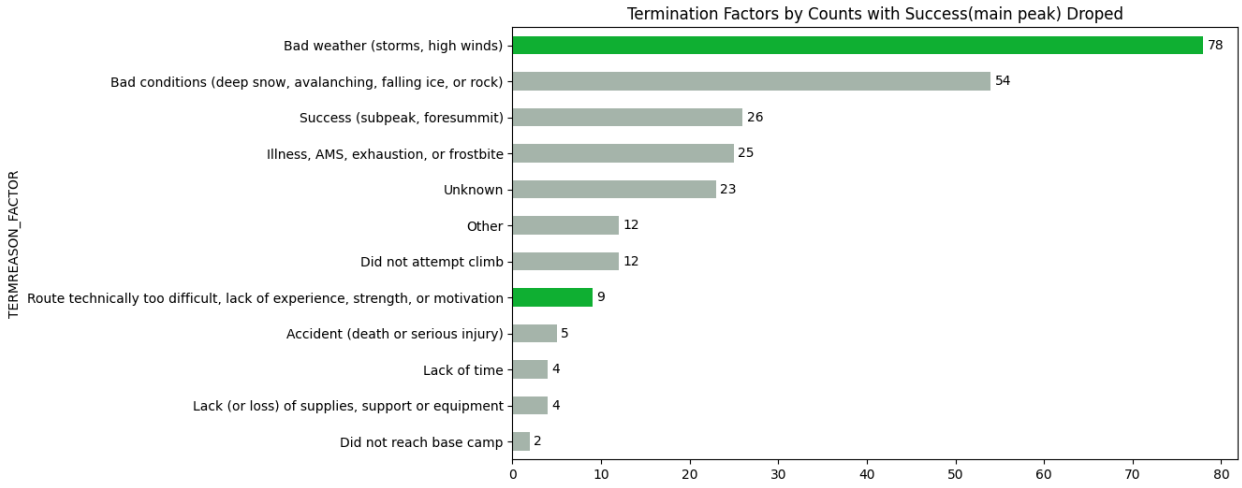
How does the use of supplemental oxygen (O2USED, O2NONE) affect summit success rates?

Since higher altitudes have lower oxygen levels, it could be helpful to carry cans of oxygen during summit attempts. There are some entries in this dataset where O2USED and O2NONE are disregarded in the following graph. There is an overall correlation between successful summit and oxygen use (dark green), while not using oxygen has a much smaller difference in success and failure.



How often does bad weather (TERMREASON = 4) play a role in termination compared to technical difficulty (TERMREASON = 10)?

Many different factors are listed in this dataset as reasons for termination of attempt . Two possible causes of termination are bad weather and technical difficulties. As seen in the graph below, bad weather is the main contributor to termination, while technical difficulties only occur in 9 cases. This implies that bad weather and conditions are the main difficulties teams face when climbing in the Himalayas.



Are expeditions with no hired personnel (NOHIRED) associated with higher or lower death rates?

Some climbers would hire persons that have more experience in the Himalayas to aid in their expedition. Knowing whether hired personnel might relate to death rates would help climbers decide whether hiring would be necessary or helpful. After conducting a Point-biserial correlation test on the data, a p value of 0.1277 and a correlation coefficient of -0.0513 is obtained. The p values are high meaning a high amount of uncertainty. Similarly the coefficient is close to zero, suggesting low correlation. This uncertainty is likely due to the overall small number of expeditions resulting in deaths in the data.

RECOMMENDATIONS

Based only on the observed associations, it is recommended to carry oxygen and avoid bad weather and conditions. Also avoid mountain ranges such as Api/Byas Risi/Guras that have a low climb success rate. By carrying oxygen for summit attempts climbers would have a higher chance in combating the low oxygen levels in high altitudes. Although the extra weight in carrying

oxygen might hinder climbing, the data supports this recommendation with high levels of confidence. By avoiding bad weather and conditions, the probability of premature termination can be significantly lowered. Less extreme weather and conditions would also result in an easier climb. While the unknown reasons of termination might affect the exact numbers, a high level of confidence is carried with this recommendation. Another way to avoid difficult terrain is by avoiding mountain ranges with high unclimb rates. However this could be affected by the fact that lower experienced climbers might choose easier climbs and result in higher chances of failure. I can conclude with moderate certainty that avoiding mountains with a high ratio of unclimb would result positively.

CITATIONS

[1] TidyTuesday, “The History of Himalayan Mountaineering Expeditions,” R4DS Online Learning Community, Jan. 21, 2025. Accessed: Aug. 21, 2026.